

THEORIA OMNIA: Emergent Spacetime Quantum-Entanglement Theory (ESQET)

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December 2025

Abstract

This paper introduces Emergent Spacetime Quantum-Entanglement Theory (ESQET), a unification framework proposing that spacetime, gravity, and coherent quantum behavior arise from dynamics in a fundamental, dimensionless Spacetime Information Field denoted $\mathcal{S}$. ESQET treats General Relativity and Quantum Mechanics as emergent phenomena generated by a single underlying information-theoretic substrate regulated by the Quantum Coherence Function ($\mathcal{F}_{QC}$). We propose four falsifiable predictions.

1 Introduction

Physics is currently bifurcated between General Relativity (GR) and Quantum Mechanics (QM). ESQET posits that gravity is not fundamental but rather the macroscopic curvature signature of coherence dynamics in a deeper information-theoretic field $\mathcal{S}$.

2 The Spacetime Information Field ($\mathcal{S}$)

We define $\mathcal{S}(x, y, z, t)$ as a dimensionless scalar field. Spacetime geometry emerges through gradients in $\mathcal{S}$.

- **Axiom 1 (Nonlocality):** $\mathcal{S}$ maintains instantaneous global coherence.
- **Axiom 2 (Gradient Stability):** Stable spacetime arises where $\nabla \mathcal{S} \rightarrow \emptyset$.
- **Axiom 3 (Curvature Encoding):**

$$G_{\mu\nu} = \alpha \partial_\mu \partial_\nu \mathcal{S} + \beta (\partial_\mu \mathcal{S})(\partial_\nu \mathcal{S})$$

3 The Quantum Coherence Function ($\mathcal{F}_{QC}$)

The core operator of ESQET is:

$$\mathcal{F}_{QC} = \oint_{\gamma} e^{i\pi\Delta\mathcal{S}} \varphi^{-\Delta\mathcal{S}} d\gamma$$

Where:

- $\pi \approx 3.14159265358979323846$ (Phase closure).
- $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618033988749895$ (Minimum-energy packing).

4 Golden Gravity

Curvature amplification occurs when the golden-ratio term dominates:

$$\varphi^{-\Delta\mathcal{S}} \rightarrow \text{Curvature Amplification Factor}$$

5 Falsifiable Predictions

1. **Entanglement-Enhanced Curvature:** Entangled systems generate measurable deviations in local gravitational potential $\Delta g \neq 0$.
2. **Scale-Dependent Gravity Drift:** Deviations in g at centimeter scales.
3. **Cosmological Coherence Polarization:** CMB signatures matching $\text{mode}_\varphi = \varphi^{-n}$.
4. **LIGO Coherence-Mode Detection:** Scalar coherence pulses detectable via phase-correlation modules.

6 Conclusion

ESQET suggests spacetime is not the stage, but the result of information density. By reframing unification through $\mathcal{S}$, we move toward an information-theoretic foundation of physical law.

References

- [1] Kauffman, S. A. (2019). *A World Beyond Physics: The Emergence and Evolution of Life*. Oxford University Press. (The Adjacent Possible).
- [2] Wheeler, J. A. (1990). *Information, Physics, Quantum: The Search for Links*.
- [3] Verlinde, E. (2011). *On the Origin of Gravity and the Laws of Newton*. Journal of High Energy Physics.